

SCALABILITY & FUTURE PLAN

Date: 25.09.2026

Team ID: SWTID-2026-5226

Project Name: FitBuddy AI Fitness Plan Generator using Gemini Models

Maximum Marks: 1 Mark

This document outlines how the FitBuddy AI solution can be scaled to handle larger user bases, increased data loads, and extended features in the future. It also provides a roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations

S.No	Limitation	Impact	Priority to Address
1	Limited number of concurrent users	May affect response time	High
2	Limited data storage	May restrict long-term user data	Medium
3	Dependence on AI model availability	May affect plan generation	Medium

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports normal user load	Use scalable server resources and load balancing
2	Data Storage	Stores essential user information	Use scalable cloud database and backup
3	Performance	AI plan generation is performed on request	Optimize API calls and use caching where suitable
4	Security	Basic application security	Strengthen authentication, authorization, and data protection

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Improve existing fitness-plan generation and user experience	Short Term	Better usability
Phase 2	Add more personalized fitness and nutrition features	Medium Term	More useful plans
Phase 3	Support larger user base and stronger data management	Long Term	Improved scalability
Phase 4	Add advanced AI-based recommendations and integrations	Future	Extended project capabilities